



Momentum and Impulse · Worksheet 1

Further Mechanics 1 Chapter 1: Momentum and Impulse. Answer all questions. Show your working clearly. Total: 66 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 1. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ ms}^{-2}$. In every collision question state clearly which direction you are taking as positive. Give impulse in N s and non-exact answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Momentum and impulse	20	
B Conservation of momentum in direct collisions	21	
C Impulse and momentum in two dimensions	20	
Roar-level challenge	5	
Total	66	

Section A Momentum and impulse

20 marks

1. A particle of mass 2.5 kg is moving in a straight line with speed 6 ms^{-1} . A constant force of 8 N then acts on the particle for 3 s, in the direction of motion. [2]
Find the magnitude of the impulse of the force and the speed of the particle at the end of the 3 s.

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2. A ball of mass 0.4 kg is moving horizontally with speed 15 ms^{-1} when it hits a vertical wall at right angles. It rebounds along the same line with speed 10 ms^{-1} . [3]
Find the magnitude of the impulse exerted by the wall on the ball.

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3. A particle of mass 2 kg is moving in a straight line with speed 5 ms^{-1} . It receives an impulse of magnitude 16 N s in the direction opposite to its motion. [3]
Find the speed of the particle immediately afterwards and state its direction of motion.

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4. A ball of mass 0.25 kg is dropped from rest from a height of 3.6 m above a horizontal floor. It rebounds vertically and rises to a height of 2.5 m . Air resistance is ignored. [4]
Find the magnitude and direction of the impulse exerted by the floor on the ball.

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5. A particle P of mass 0.5 kg is moving in a straight line on a smooth horizontal surface with speed 4 ms^{-1} . It receives an impulse of magnitude $J\text{ N s}$, after which it moves in the opposite direction with speed 6 ms^{-1} .

(a) Find the value of J .

[3]

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- (b) After receiving the impulse, P moves onto a rough horizontal surface and comes to rest 2.5 s later. Use the impulse–momentum principle to find the frictional force on P , and hence find the coefficient of friction between P and the rough surface.

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(c) State one modelling assumption you have made in part (b).

[1]

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(Total for Question 5 is 8 marks)

**Section B Conservation of momentum in direct collisions**

21 marks

6. A particle A of mass 4 kg , moving at 6 ms^{-1} on a smooth horizontal surface, collides directly with a particle B of mass 2 kg which is at rest. The particles coalesce. [2]
Find the speed of the combined particle immediately after the collision.

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7. Particles A of mass 3 kg and B of mass 2 kg move towards each other along the same straight line on a smooth surface, with speeds 5 ms^{-1} and 2 ms^{-1} respectively, and collide directly. Immediately after the collision A is still moving in the same direction, with speed 1 ms^{-1} . [3]



Find the speed of B immediately after the collision and state its direction of motion.

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8. A gun of mass 1.2 kg is free to move horizontally. It fires a bullet of mass 0.02 kg horizontally with speed 300 ms^{-1} . [3]
Find the speed with which the gun recoils, and the magnitude of the impulse exerted on the bullet by the gun.

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9. Two particles A and B , of masses 2 kg and 3 kg, lie at rest on a smooth horizontal table, connected by a light inextensible string which is slack. A is projected directly away from B with speed 10 ms^{-1} . Find the common speed of the particles immediately after the string becomes taut, and the magnitude of the impulse in the string when it jerks tight. [4]

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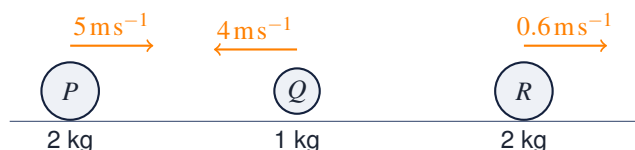
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10. Three particles P , Q and R , of masses 2 kg, 1 kg and 2 kg respectively, lie in a straight line on a smooth horizontal surface, with Q between P and R . P and Q move towards each other with speeds 5 m s^{-1} and 4 m s^{-1} respectively, and R moves away from Q with speed 0.6 m s^{-1} .



After P and Q collide, the direction of motion of P is unchanged, and the speed of Q is three times the speed of P .

- (a) Find the speeds of P and Q immediately after this collision. [4]

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- (b) Find the magnitude of the impulse exerted by P on Q in the collision. [2]

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- (c) Q then collides with R and the two particles coalesce. Find the speed of the combined particle, and determine whether P collides with it again. [3]

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(Total for Question 10 is 9 marks)

**Section C Impulse and momentum in two dimensions**

20 marks

- 11.** A particle of mass 2 kg is moving with velocity $(3\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$. [2]

Find the momentum of the particle as a vector, and the magnitude of the momentum.

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- 12.** A particle of mass 0.5 kg is moving with velocity $(4\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$ when it receives an impulse of $(3\mathbf{i} + 5\mathbf{j}) \text{ N s}$. [3]

Find the velocity of the particle immediately after the impulse, and its speed.

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- 13.** A particle of mass 3 kg is moving with velocity $(2\mathbf{i} + 5\mathbf{j}) \text{ m s}^{-1}$. It receives an impulse $\mathbf{I} \text{ N s}$, after which its velocity is $(-4\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$. [3]

Find $\mathbf{I}$ and its magnitude.

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- 14.** A ball of mass 0.2 kg is moving horizontally with velocity $15\mathbf{i} \text{ m s}^{-1}$ when it is hit by a bat. Immediately afterwards the ball has velocity $20\mathbf{j} \text{ m s}^{-1}$. [4]

Find the magnitude of the impulse exerted by the bat on the ball, and the angle between the impulse and the original direction of motion of the ball.

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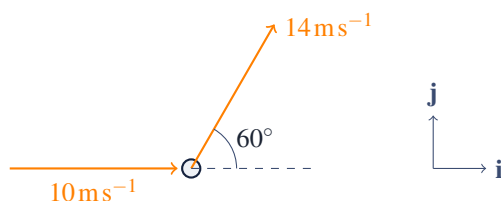
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15. A ball of mass 0.4 kg is moving on a smooth horizontal surface with velocity $10\mathbf{i} \text{ ms}^{-1}$. It is struck by a club, which gives it an impulse $\mathbf{J} \text{ N s}$. Immediately after the impulse, the ball moves with speed 14 ms^{-1} in a direction which makes an angle of 60° with $\mathbf{i}$, as shown.



- (a) Write down the velocity of the ball immediately after the impulse in the form $(a\mathbf{i} + b\mathbf{j}) \text{ ms}^{-1}$, giving exact values of a and b . [2]

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- (b) Find $\mathbf{J}$, and show that the magnitude of $\mathbf{J}$ is 5.00 N s to 3 significant figures. [3]

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- (c) Find the angle between $\mathbf{J}$ and the original direction of motion of the ball. [3]

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(Total for Question 15 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Two particles A and B , of masses 2 kg and 3 kg, are attached to the ends of a light inextensible string which passes over a smooth fixed pulley. A rests on a horizontal floor. B is held at rest with the string slack, and is then released, so that it falls freely. B has fallen 0.9 m when the string becomes taut and A is jerked into motion. [5]
- Find the common speed of A and B immediately after the string becomes taut, and the magnitude of the impulse in the string.

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END OF WORKSHEET

Total: 66 marks